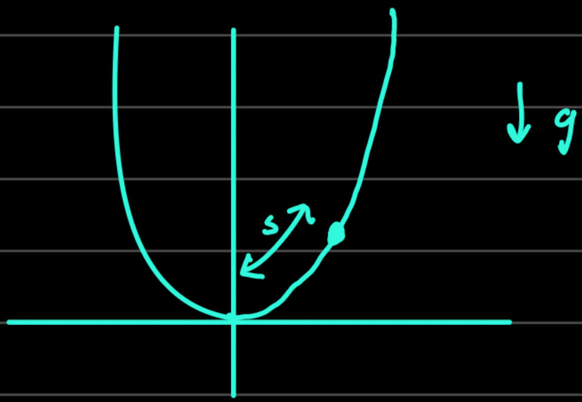
1. *Small oscillations along a curve:* A particle of mass m moves in a constant vertical gravitational field along the curve defined by $y = a\,x^4$, where y is the vertical direction and $a > 0$. Find the equation of motion for small oscillations about the position of equilibrium.



$$y = ax^4$$
$$\dot{y} = 4ax^3\dot{x}$$

$$L = \frac{1}{2}m\dot{x}^2 + \frac{1}{2}m\cdot 16a^2x^6\dot{x}^2 - mg(ax^4)$$

$$L = \frac{1}{2}m\dot{x}^2(1 + 16a^2x^6) - mgax^4$$

$$\frac{\partial L}{\partial \dot{x}} = m\dot{x}(1+16a^2x^6) \qquad \frac{\partial L}{\partial x} = \frac{1}{2}m\dot{x}^2\cdot 96a^2x^5 - 4mgax^3$$

$$\Rightarrow \quad m\ddot{x}(1+16a^2x^6) + m\dot{x}^2(96a^2x^5) = \frac{1}{2}m\dot{x}^2(96ax^5) - 4mgax^3$$

$$\Rightarrow \quad m\ddot{x} = -4mgax^3$$

$$\boxed{\ddot{x} = -4gax^3}$$

2. *System with two degrees of freedom:* Determine the oscillations of a system with two degrees of freedom (say, x and y) whose Lagrangian is given by

m=1 here

$$L = \frac{1}{2}(\dot{x}^2 + \dot{y}^2) - \frac{{\omega_0}^2}{2}(x^2 + y^2) + \alpha\, x\, y.$$

From Morin, but is a clearer way.

$$\frac{\partial L}{\partial \dot{x}} = \dot{x} \qquad \frac{\partial L}{\partial x} = -\omega_0^2 x + \alpha y$$

$$\frac{\partial L}{\partial \dot{y}} = \dot{y} \qquad \frac{\partial L}{\partial y} = -\omega_0^2 y + \alpha x$$

$$\Rightarrow \quad \ddot{x} = -\omega_0^2 x + \alpha y \qquad \text{Put } \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} A \\ B \end{pmatrix} e^{i\gamma t}$$

$$\ddot{y} = -\omega_0^2 y + \alpha x$$

$$\Rightarrow -A\gamma^2 = -A\omega_0^2 + B\alpha$$

$$-B\gamma^2 = -B\omega_0^2 + A\alpha$$

$$\Rightarrow -A\gamma^2 + A\omega_0^2 - B\alpha = 0$$

$$-A\alpha - B\gamma^2 + B\omega_0^2 = 0$$

or

$$\underbrace{\begin{pmatrix} -\gamma^2 + \omega_0^2 & -\alpha \\ -\alpha & -\gamma^2 + \omega_0^2 \end{pmatrix}}_{z} \begin{pmatrix} A \\ B \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

A, B nontrivially zero when $\det z = 0$

$$\Rightarrow (\gamma^2 - \omega_0^2)^2 = \alpha^2$$

$$\gamma^2 - \omega_0^2 = \alpha \qquad\qquad \gamma^2 - \omega_0^2 = -\alpha$$

$$\gamma^2 = \alpha + \omega_0^2 \qquad\qquad \gamma^2 = -\alpha + \omega_0^2$$

$$\gamma = \underbrace{\sqrt{\alpha + \omega_0^2}}_{1}, \underbrace{-\sqrt{\alpha + \omega_0^2}}_{2} \qquad\qquad \gamma = \underbrace{\sqrt{-\alpha + \omega_0^2}}_{3}, \underbrace{-\sqrt{-\alpha + \omega_0^2}}_{4}$$

So for 1 and 2, $\gamma^2 - \omega_0^2 = \alpha$

$$\Rightarrow \begin{pmatrix} -\alpha & -\alpha \\ -\alpha & -\alpha \end{pmatrix} \begin{pmatrix} A \\ B \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad \Rightarrow \quad A = -B$$

For 3 and 4, $\gamma^2 - \omega_0^2 = -\alpha$,

$$\begin{pmatrix} \alpha & -\alpha \\ -\alpha & \alpha \end{pmatrix} \begin{pmatrix} A \\ B \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow \quad A = B$$

So
$$\boxed{\begin{aligned} \begin{pmatrix} x \\ y \end{pmatrix} = {} & A_1 \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{i\sqrt{\alpha+\omega_0^2}\,t} + A_2 \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{-i\sqrt{\alpha+\omega_0^2}\,t} + A_3 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{i\sqrt{-\alpha+\omega_0^2}\,t} \\ & + A_4 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{-i\sqrt{-\alpha+\omega_0^2}\,t} \end{aligned}}$$

So $x+y$ is the normal coordinate associated with the normal mode $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$

and $x-y$ is the normal coordinate associated with the normal mode $\begin{pmatrix} 1 \\ -1 \end{pmatrix}$

because $x+y$ eliminates oscillation due to the second frequency and $x-y$ eliminates oscillation due to the first frequency (and not because $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ are coefficients in $x+y$ and $\begin{pmatrix} 1 \\ -1 \end{pmatrix}$ are coefficients of $x-y$).

So if you had $\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 3 \\ 2 \end{pmatrix} \cos\omega_1 t + \begin{pmatrix} 1 \\ -5 \end{pmatrix} \cos\omega_2 t$

Then $\left\{ \begin{array}{l} 5x+y \text{ eliminates } \omega_2 \text{ oscillation and} \\ 2x-3y \text{ eliminates } \omega_1 \text{ oscillation, so these are the normal coordinates.} \end{array} \right.$

3. *Space oscillator:* Consider a particle moving in the central potential $U(r) = k\,r^2/2$. Determine the trajectory of the particle in its plane of motion.

$$\mathcal{L} = \frac{1}{2}m\dot{x}^2 + \frac{1}{2}m\dot{y}^2 - \frac{kx^2}{2} - \frac{ky^2}{2}$$

$$\Rightarrow \quad x = A\cos(\omega t + \alpha) \quad , \quad y = B\cos(\omega t + \beta)$$

or let relative phase by δ

$$x = A\cos\omega t \qquad y = B\cos(\omega t + \delta)$$

$$x = A\cos\omega t \qquad y = B\cos\delta\cos\omega t - B\sin\delta\sin\omega t$$

$$\Rightarrow y = \frac{B\cos\delta}{A}x - \frac{B\sin\delta}{A}\sqrt{A^2 - x^2}$$

$$\frac{B}{A}\sin\delta\sqrt{A^2 - x^2} = \frac{B\cos\delta}{A}x - y$$

$$B^2\sin^2\delta - \frac{B^2}{A^2}\sin^2\delta\, x^2 = \frac{B^2}{A^2}\cos^2\delta\, x^2 - 2xy\frac{B}{A}\cos\delta + y^2$$

$$B^2\sin^2\delta = \frac{B^2}{A^2}x^2 - 2xy\frac{B}{A}\cos\delta + y^2$$

$$\boxed{\sin^2\delta = \frac{x^2}{A^2} + \frac{y^2}{B^2} - \frac{2xy}{AB}\cos\delta}$$

when $\delta = 0, \pi$: $\quad 0 = \left(\frac{x}{A} \pm \frac{y}{B}\right)^2 \Rightarrow \quad y = \pm\frac{B}{A}x$

when $\delta = \pi/2$: $\quad 1 = \frac{x^2}{A^2} + \frac{y^2}{B^2} \Rightarrow$ standard ellipse.

5. *Forced oscillations in the presence of friction:* Determine the forced oscillations due to the external force $F = F_0 \exp(\alpha t) \cos(\gamma t)$ in the presence of friction.

$$\ddot{x} + 2\lambda\dot{x} + \omega^2 x = \frac{F_0}{m} e^{(\alpha+i\gamma)t}$$

Put $x = b e^{(\alpha+i\gamma)t}$

$$\Rightarrow \quad b(\alpha+i\gamma)^2 + 2\lambda b(\alpha+i\gamma) + \omega^2 b = \frac{F_0}{m}$$

$$\Rightarrow \quad b = \frac{F_0}{m}\left(\frac{1}{(\alpha+i\gamma)^2 + 2\lambda(\alpha+i\gamma) + \omega^2}\right)$$

$$b = \frac{F_0}{m}\left(\frac{1}{\alpha^2 - \gamma^2 + 2\alpha\gamma i + 2\lambda\alpha + 2\lambda\gamma i + \omega^2}\right)$$

$$b = \frac{F_0}{m}\left(\frac{1}{(\alpha^2 - \gamma^2 + 2\lambda\alpha + \omega^2) + i(2\alpha\gamma + 2\lambda\gamma)}\right)$$

$$\Rightarrow b = \frac{F_0}{m}\left(\frac{\alpha^2 - \gamma^2 + 2\lambda\alpha + \omega^2 - i(2\alpha\gamma + 2\lambda\gamma)}{(\alpha^2 - \gamma^2 + 2\lambda\alpha + \omega^2)^2 + 4\gamma^2(\alpha+\lambda)^2}\right)$$

$$= Be^{i\delta}$$

Then $$B = \frac{F_0}{m} \frac{1}{\sqrt{(\alpha^2 - \gamma^2 + 2\lambda\alpha + \omega^2)^2 + 4\gamma^2(\alpha+\lambda)^2}}$$

$$\tan\delta = \frac{-2\gamma(\alpha+\lambda)}{\alpha^2 - \gamma^2 + 2\lambda\alpha + \omega^2}$$